

Nielsen Norman Group

Heuristic Evaluation

Workbook

Use this workbook to conduct your own heuristic evaluation.

For each of Jakob's 10 Usability Heuristics, look for specific places where the interface fails to adhere to the guideline. Write your recommendations for how to fix those usability issues.

<https://www.nngroup.com/articles/ten-usability-heuristics/>

Nielsen Norman Group

Heuristic Evaluation

Workbook

Evaluator:

Date:

Product:

Task:

1

Visibility of System Status

The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time.

- Does the design clearly communicate its state? Is feedback presented quickly after user actions?
-

Issues

PawTrack generally communicated system status well throughout the prototype. Users received confirmation messages after successfully booking an appointment, and vaccination status indicators clearly showed whether a pet's records were up to date or overdue. The active navigation tab was visually highlighted so users always knew which section they were in. However, the prototype lacked loading indicators or progress feedback during transitions between screens, which left some users momentarily uncertain whether their action had been registered or if the app had frozen.

Recommendations

Add visible loading spinners or progress bars during screen transitions and after form submissions so users are never left guessing whether the system is processing their action. Appointment booking confirmations should also include a timestamp and summary card to reinforce that the action was completed successfully.

2

Match Between System and the Real World

The design should speak the users' language. Use words, phrases, and concepts familiar to the user, rather than internal jargon. Follow real-world conventions, making information appear in a natural and logical order.

Will user be familiar with the terminology used in the design? Do the design's

- controls follow real-world
- conventions?

Issues

Most of the language used in PawTrack was straightforward and pet-owner friendly, using terms like "My Pets," "Book Appointment," and "Vaccination Records" that users found intuitive. However, some medical terminology within the vaccination records section — such as vaccine strain names and dosage classifications — was unfamiliar to non-veterinary users. A few participants were also confused by the label "Health Logs" as they expected it to be called "Medical History," which is a more commonly understood term among pet owners.

Recommendations

Replace technical veterinary labels with plain, everyday language wherever possible. For example, rename "Health Logs" to "Medical History" and provide brief tooltip descriptions beside unfamiliar vaccine names so users can understand the information without needing a veterinary background. Icons should also follow real-world conventions, such as using a calendar icon for appointments and a syringe icon for vaccination records.

Heuristic Evaluation Workbook

Issues

PawTrack allowed users to navigate back to previous screens using the back arrow, and a cancel option was available when booking or canceling an appointment. However, the prototype did not support undo or redo functionality. Once a user confirmed a booking or submitted pet profile information, there was no way to reverse or edit the entry without going through the entire process again from the beginning. Some users also noted that the cancel button during the booking flow was small and easy to miss.

Recommendations

Implement undo functionality so users can reverse recently submitted actions, particularly for appointment bookings and pet profile edits. Make the cancel button more visually prominent by increasing its size and placing it in a consistent and accessible location within each screen. An editable confirmation screen before final submission would also give users a clear last opportunity to review and correct their input.

User Control and Freedom

Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted action without having to go through an extended process.

Does the design allow users to go back a step in the process?

Are exit links easily discoverable?

Issues

PawTrack maintained a generally consistent visual style throughout its screens, using the same color palette, button styles, typography, and iconography across the pet profile, booking, and vaccination record sections. Navigation followed standard mobile conventions with a bottom tab bar, which users found familiar and easy to use. Minor inconsistencies were observed, however, in the way action buttons were labeled — for instance, "Save" was used on some screens while "Confirm" was used on others for what appeared to be the same type of action.

Recommendations

Standardize all action button labels across every screen so that identical actions always use the same terminology. Conduct a full interface audit to ensure that button sizes, spacing, font weights, and icon styles remain uniform throughout the app. Maintaining this consistency will reduce user confusion and make the interface feel more polished and professional.

Consistency and Standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions.

Does the design follow industry conventions?

Are visual treatments used consistently throughout the

design?

Heuristic Evaluation Workbook



Error Prevention

Good error messages are important, but the best designs carefully prevent problems from occurring in the first place. Either eliminate error-prone conditions, or check for them and present users with a confirmation option before they commit to the action.

- Does the design prevent slips by using helpful constraints?
- Does the design warn users before they perform risky actions?

Issues

PawTrack included a confirmation dialog when users attempted to cancel an existing appointment, which effectively prevented accidental cancellations. The booking form also restricted users from selecting past dates for appointments, which reduced date entry errors. However, the pet profile registration form did not include real-time input validation, meaning users could submit incomplete or incorrectly formatted information — such as leaving the pet's age field blank or entering letters in a numeric field — without being alerted until after submission.

Recommendations

Introduce real-time inline validation for all form fields in PawTrack so users are notified of errors as they type rather than after submission. Required fields should be clearly marked with an asterisk or highlighted border. Additionally, a final review screen before confirming any booking or profile submission would allow users to catch and correct mistakes before they are committed to the system.

6

Recognition Rather Than Recall

Minimize the user's memory load by making elements, actions, and options visible. The user should not have to remember information from one part of the interface to another. Information required to use the design (e.g. field labels or menu items) should be visible or easily retrievable when needed.

- Does the design keep important information visible, so that users do not have to memorize it?
- Does the design offer help in-context?

Issues

PawTrack displayed pet names, profile photos, and vaccination status prominently on the home dashboard, making it easy for users to recognize key information at a glance without needing to memorize details. Appointment schedules and upcoming vaccine due dates were also visible within the main screens. However, one recurring issue was that the "Add Booking" button was not immediately noticeable to several users during testing because it blended into the interface and lacked a clear distinguishing label. Users only located it after being pointed to it by the evaluator.

Recommendations

Redesign the "Add Booking" button to be more visually prominent — using a contrasting color, larger size, and a clear text label alongside the icon — so users can locate and understand its function at a glance. Persistent labels beneath all navigation icons would also help users identify each section without needing to guess or recall icon meanings from prior use.

Heuristic Evaluation Workbook



Flexibility and Efficiency of Use

Shortcuts — hidden from novice users — may speed up the interaction for the expert user such that the design can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

- Does the design provide accelerators like keyboard shortcuts and touch gestures?
- Is content and functionality personalized or customized for individual users?



Aesthetic and Minimalist Design

Interfaces should not contain information that is irrelevant or rarely needed. Every extra unit of information in an interface competes with the relevant units of information and diminishes their relative visibility.

- Is the visual design and content focused on the essentials?
- Have all distracting, unnecessary elements been removed?

Issues

PawTrack supported standard mobile touch gestures such as tapping and scrolling, which allowed users to navigate the app efficiently. The app also offered a degree of personalization through individual pet profiles, each displaying that specific pet's vaccination history and upcoming appointments separately. However, the prototype did not include any shortcut features such as quick-book options for frequently visited veterinary clinics, or swipe gestures for common actions like dismissing notifications or switching between pet profiles.

Recommendations

In future versions of PawTrack, consider adding a quick-booking shortcut on the home dashboard that allows returning users to rebook a recent appointment with a single tap. Swipe gestures for switching between pet profiles and dismissing alerts would also improve efficiency for experienced users. A "Favorites" feature for frequently visited veterinary clinics could further personalize and speed up the booking experience.

Issues

PawTrack presented a clean and visually appealing interface that prioritized essential pet care information. Screens were free from excessive text, unnecessary graphics, and unrelated content, making it easy for users to focus on the core tasks of managing pet profiles, viewing records, and booking appointments. The color scheme and layout were well-organized and contributed to a comfortable browsing experience. A minor concern was observed on the home dashboard, where the simultaneous display of multiple pet cards, upcoming appointments, and vaccination alerts in a single view felt slightly crowded on smaller screen sizes.

Recommendations

Maintain PawTrack's clean and minimal design approach as a core design principle going forward. For the home dashboard, consider implementing a collapsible or tabbed layout that allows users to switch between pet summaries, appointment schedules, and health alerts separately, reducing visual density without hiding important information. Consistent use of white space between content sections would also improve overall readability and reduce cognitive load.

Heuristic Evaluation Workbook

Help Users Recognize, Diagnose, and Recover from Errors

Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution.

- Does the design use traditional error message visuals, like bold, red text?
- Does the design offer a solution that solves the error immediately?

Issues

PawTrack's prototype included basic confirmation and cancellation prompts that helped prevent unintended actions such as accidental appointment cancellations. However, the current prototype lacked specific and descriptive error messages. When a user attempted to submit an incomplete booking form, no clear error indicator appeared to identify which field was missing or what correction was needed. The prototype was primarily focused on demonstrating intended workflows rather than fully handling error states, which left users uncertain about what went wrong and how to fix it.

Recommendations

Future versions of PawTrack should include clearly worded inline error messages that appear directly below the problematic field in red text, specifying exactly what is wrong — for example, "Please enter your pet's date of birth" or "This time slot is already fully booked. Please select a different time." A summary error banner at the top of the form listing all incomplete fields would further assist users in correcting their submissions quickly and confidently.

10 Help and

Documentation

It's best if the system doesn't need any additional explanation.

However, it may be necessary to provide documentation to help users understand how to complete their tasks.

- Is help documentation easy to search?
- Is help provided in context right at the moment when the user requires it?

Issues

The current PawTrack prototype did not include any built-in help documentation, onboarding tutorial, or searchable FAQ section. Since the prototype was designed to demonstrate core features and interface flow, help components were not part of the initial build. Despite this absence, most participants were able to complete the assigned tasks independently, suggesting that the interface was largely intuitive. However, first-time users unfamiliar with the app's structure — particularly the vaccination record section and the booking flow — occasionally expressed uncertainty about where to begin or what a particular screen was asking them to do.

Recommendations

Future versions of PawTrack should include a brief onboarding walkthrough for first-time users that introduces the main features — pet profile creation, vaccination tracking, and appointment booking — with illustrated step-by-step guidance. Contextual help tooltips placed near complex fields or unfamiliar icons would provide in-the-moment support without requiring users to leave the screen. A searchable FAQ or Help Center accessible from the settings menu would also serve as a reliable fallback for users who encounter difficulty during independent use.